

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

4 Formal Models of Language (pjb48)

regular languages (a) The following is a pattern for some legal strings in a language:

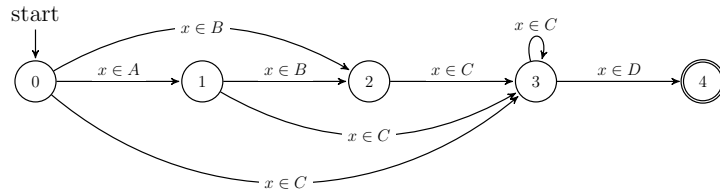
$$[a \in A]\{0,1\} [b \in B]\{0,1\} [c \in C]\{1,n\} [d \in D]\{1,1\}$$

where A is a finite set of characters from the alphabet, Σ ; similarly for B, C, D . The sets A, B, C and D are disjoint. $\{x, y\}$ indicates that the previous bracket must match at least x times but no more than y times.

DFA

(i) Specify a Deterministic Finite Automaton, M_1 , that can recognise these strings only. [4 marks]

Answer: $M_1 = (\mathcal{Q}, \Sigma, \Delta, s, \mathcal{F})$.



where:

- $\Delta \subseteq \mathcal{Q} \times \Sigma \times \mathcal{Q}$ as shown in a transition table or diagram (partial or total function accepted) [4 marks — 1 for accounting for each part of the regexp]
 - $\mathcal{Q} = \{0, 1, 2, 3, 4\}$
 - $\Sigma = A \cup B \cup C \cup D$
 - $s = 0$
 - $\mathcal{F} = \{4\}$
-

regular grammar

(ii) Design a Regular Grammar, G_1 , which generates $L(M_1)$. [4 marks]

Answer: $G_1 = (\mathcal{N}, \Sigma, S, \mathcal{P})$

- $\mathcal{N} = \{S, X, Y, Z\}$
- $\Sigma = A \cup B \cup C \cup D$
- $S = S$
- $\mathcal{P} = \{$
 $S \rightarrow x \in A \ X | x \in B \ Y | x \in C \ Z$
 $X \rightarrow x \in B \ Y | x \in C \ Z$
 $Y \rightarrow x \in C \ Z$
 $Z \rightarrow x \in C \ Z | x \in D\}$

[1 mark for using left or right RG rules only, 1 mark for a grammar that handles the loop, 1 mark for end states and 1 mark for other transitions]

human language

(iii) Describe a set of strings in a natural language that could be generated by

G_1 given an appropriate Σ and its subsets A , B , C and D . [1 mark]

Answer: There are many. In English this could describe the language of proper names: $A = \{Prof|Dr\}$, $B = \{Sir|Dame\}$, $C = \{Forenames\}$, $D = \{Surname\}$; but any description of appropriately co-ordinated modifiers would score a point.

context free
languages

- (b) We can hypothesise that matches of the following pattern are always valid constructions in English:

[The *Noun*]{n,n} [*Verb*]{n,n}

where *Noun* represents the coordinated members of a finite set; similarly for *Verb*.

natural language

- (i) Now consider the following English sentence which matches the pattern when $n = 1$:

The vaccine worked

Provide example sentences that extend this sentence for the case when $n = 2$ and $n = 3$. [2 marks]

Answer: Examples would be:

The vaccine the lab made worked

The vaccine the lab the government funded made worked

[1 mark per example]

closure properties

- (ii) Assuming that these constructions are part of the English language, would this mean that English is a Context-Free Language? Justify your answer. [3 marks]

Answer: We can show that these constructions are not regular by the pumping lemma for regular languages but this does not prove they are context free [1 mark]. The complexity of a sub-language is not necessarily the complexity of a language [1 mark]. We can use the knowledge that regular languages are closed under intersection to show by contradiction that an assumption that English is regular must be wrong [1 mark].

CNF

- (iii) Design a grammar in Chomsky Normal Form, G_2 , which generates the finite matches of the pattern. [3 marks]

Answer: $G_2 = (\mathcal{N}, \Sigma, S, \mathcal{P})$

- $\mathcal{N} = \{S, A, X, D, N, V\}$
- $\Sigma = Det \cup Noun \cup Verb$
- $S = S$

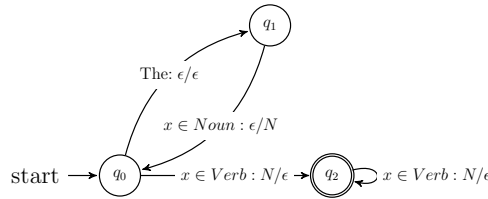
- $\mathcal{P} = \{$
 $S \rightarrow A X | A V,$
 $X \rightarrow S V,$
 $A \rightarrow D N,$
 $D \rightarrow \text{the},$
 $N \rightarrow x \in \text{Noun}$
 $V \rightarrow x \in \text{Verb}\}$

[1 mark for CNF, 1 mark for same number of Ns and Vs, 1 mark for terminals]

PDA

(iv) Specify a Push Down Automaton, M_2 , that recognises $L(G_2)$. [3 marks]

Answer: $M_2 = (\mathcal{Q}, \Sigma, \Gamma, \Delta, s, \perp, \mathcal{F})$



where:

- $\mathcal{Q} = \{q_0, q_1, q_2\}$
- $\Sigma = \text{Det} \cup \text{Forename} \cup \text{Surname}$
- $\Gamma = \{N\}$
- $\Delta \subseteq (\mathcal{Q} \times (\Sigma \cup \epsilon) \times \Gamma) \times (\mathcal{Q} \times \Gamma^*)$ as shown in transition tables or a diagram
- $s = S$
- $\perp$ — n.b. this machine accepts on empty stack plus finishing state
- $\mathcal{F} = \{q_2\}$

[1 mark for push, 1 mark for pop, 1 mark for same number of Ns and Vs]
